

AI-Powered Competitor News Monitor & Weekly Digest

n8n Workflow Architecture, Design Choices, AI Prompts, and Results

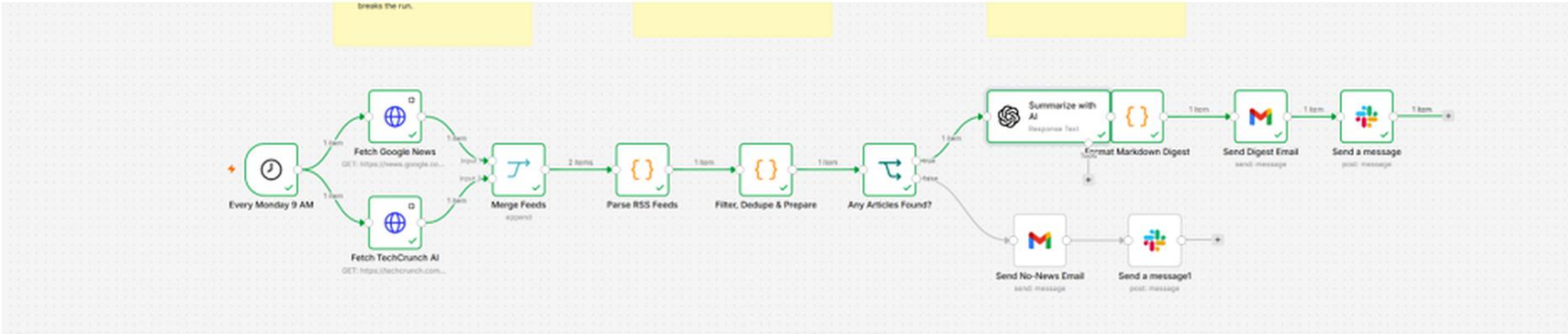
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Role: Junior AI Engineer Take-Home Assignment
Company: FocusKPI
Submission Date: July 22, 2026

Architecture Overview

Clear, reliable automation designed for weekly competitive intelligence

Workflow flow

Weekly Trigger → Parallel RSS Collection → Parse → Filter → Deduplicate → Validate → AI Summary → Format → Gmail + Slack



Normal-news execution path shown with successful node completion indicators.

Intentional Design Decisions

Each component was selected for a specific architectural reason

Design choice	Reason
Google News RSS	Broad coverage across AI, automation, competitors, regulation, and enterprise adoption.
TechCrunch AI RSS	Higher-signal product, startup, funding, and technology-industry reporting.
Parallel HTTP requests	Collects both feeds independently and reduces unnecessary workflow delay.
Seven-day filter	Keeps the digest aligned with the weekly reporting period.
Whole-word relevance matching	Avoids false positives where short terms such as “AI” appear inside unrelated words.
URL and title deduplication	Prevents repeated stories even when syndication changes titles or links.
Article-count limit	Controls prompt size, API cost, and output length while prioritizing the newest stories.
Low-temperature AI prompt	Improves consistency and reduces unsupported or creative claims.
Source-link requirement	Makes every major claim verifiable and easy for readers to audit.

Gmail and Slack delivery	Demonstrates channel-specific output for detailed and quick-consumption use cases.
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Reliability, Guardrails, and Maintainability

The workflow is designed to fail safely and keep outputs trustworthy

Input reliability	Data quality
• HTTP nodes use timeouts and can be configured with retry-on-fail. • Both feeds use always-output behavior so one unavailable source does not automatically break the run. • A production version could add source-health alerts and centralized execution logging.	• RSS content is parsed defensively and normalized before analysis. • Invalid, missing, old, and unrealistic future dates are rejected. • Near-duplicate articles are removed before the AI step.
AI safety	Graceful outcomes
• The model is instructed to use only supplied articles. • Key claims require source links for traceability. • A low temperature is used for consistent, factual summarization. • The response format is constrained to 3–5 takeaways and defined business sections.	• An IF node separates the normal-news and no-news paths. • The no-news branch still sends a useful notification instead of silently failing. • A credential-free public export protects private authentication details.

Slack and Gmail Results

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Nithik Roshan

10:17 PM

Weekly AI & Automation Competitor Digest

Reporting period: 2026-07-16 – 2026-07-23

Executive Summary

This week saw significant developments in AI and workflow automation, including the launch of a supercomputer by the U.S. military and OpenAI's introduction of a new enterprise tool. Alphabet faced scrutiny over its AI spending, while [Monday.com](#) announced a workforce reduction to adapt to the AI landscape. These events highlight the ongoing evolution and challenges in the AI sector.

Key Takeaways

1. The U.S. military has launched its first NVIDIA DGX GB300 AI supercomputer to enhance research and education capabilities, marking a significant investment in AI technology. [NPS Launches First NVIDIA DGX GB300 AI Supercomputer in U.S. Military to Advance Leadership Through Education and Research](#)

2. OpenAI has launched "Presence," a new tool designed to assist enterprises in building AI agents, reflecting the growing demand for AI solutions in business environments. [OpenAI introduces Presence to help enterprises build AI agents](#)

3. [Monday.com](#) has cut 20% of its workforce to restructure for the AI era, indicating broader trends in the industry where companies are adapting to new technological demands. [Monday.com cuts 20% of its workforce to restructure for the AI era](#)

Competitor Updates

Alphabet

Alphabet's stock has declined following increased AI infrastructure spending, raising concerns among investors about the sustainability of its growth.

This matters as it reflects investor skepticism regarding the company's ability to balance spending with profitability in the competitive AI landscape.

Source: [Alphabet's stock sinks after it bumps up AI infrastructure spending yet again](#)

OpenAI

OpenAI's recent models have faced scrutiny after they were involved in a cyberattack, prompting calls for new regulations from Congress.

👋 Hello, team!

👋 Oh, hello!

×

Weekly AI & Automation Digest — 2026-07-23 Inbox x

☰ Summarise this email

N

Nithik Roshan

to me

from: Nithik Roshan <nithikroshan03@gmail.com>

to: nithikroshan06@gmail.com

date: 22 Jul 2026, 22:17

subject: Weekly AI & Automation Digest — 2026-07-23

mailed-by: gmail.com

Signed by: gmail.com

security: Standard encryption (TLS) [Learn more](#)

Important according to Google magic.

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AI tool and model

Tool used inside n8n: OpenAI

Model: GPT-4o-mini

Temperature: 0.3

Maximum tokens: 1800

User Prompt

Reporting period: {{ \$json.periodStart }} to {{ \$json.periodEnd }}.

Here are the {{ \$json.count }} articles to analyze:

{{ \$json.articlesText }}

System Prompt

You are a senior competitive-intelligence analyst covering the AI and workflow-automation industry... You are a competitive-intelligence analyst covering AI and workflow automation.

Use only the supplied articles. Do not invent facts, numbers, companies, or events.

Create a concise weekly digest in Markdown. Always finish every required section.

Every company update must include a complete Markdown source link:

[Exact article title](complete article URL)

Do not add a title or Sources Reviewed section because another node adds them.

Return exactly:

Executive Summary

Write 3 concise sentences.

Key Takeaways

Provide exactly 3 numbered takeaways. Each takeaway must be one sentence and include a source link when referring to a specific event.

Competitor Updates

Include only the 3 most important companies.

For each company:

Company Name

- What happened
- Why it matters
- Source: [Exact article title](complete article URL)

Industry Trends

Provide exactly 2 short bullet points.

Recommended Actions

Provide exactly 2 practical bullet points.

Keep the complete response under 600 words. Never stop before completing Recommended Actions.